

AI Data Center Sustainability: Scenario Screening Across Bottlenecks, with a Focus on Photonic Interconnects

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Motivation: AI data centers face orthogonal constraints that vary by workload and site: power delivery limits, cooling/thermal limits, carbon and water limits, grid interconnection timelines, and reliability requirements. Because the dominant bottleneck is scenario-dependent, an early-stage assessment should screen all levers, then focus on deeper modeling where it is most relevant. This brief provides (1) a bottleneck/lever map and (2) a transparent, baseline-free assessment of one candidate lever: **photonic interconnects as an indirect scaling gain** that can reduce required GPU capacity, facility MW, and CO₂, *if* the workload is communication/coordination constrained.

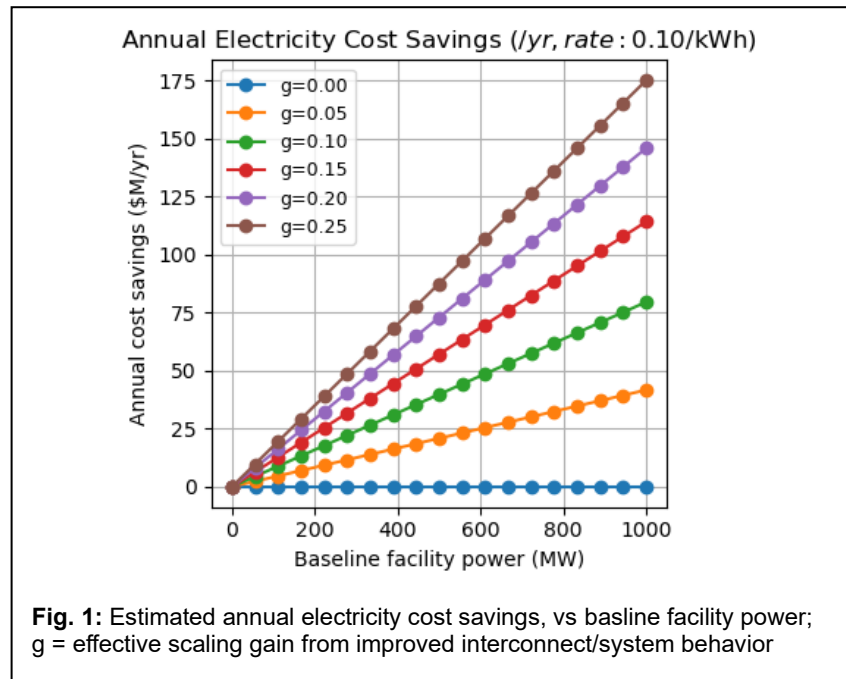
AI Datacenter Bottlenecks and Levers:

Usage scenario		Bottlenecks	Primary Levers	Decision Questions
(1)	Large distributed training (“AI factory”)	Fabric/collectives, contention, multi-tenancy interference, power cap	Interconnect scaling, scheduling/placement, utilization recovery	Are collectives/comm a large fraction of step time? Do co-running jobs interfere? Do GPUs idle waiting on sync?
(2)	Inference/serving (latency + availability)	Latency SLOs, reliability, demand spikes	Model routing/caching/batching, power management, redundancy	Is this latency- or throughput-bound? What are tail-latency drivers?
(3)	High-density racks/hot climate/water limited	Thermal & water constraints, peak ambient	Cooling architecture, setpoints, loop materials, heat rejection strategy	Is water constrained? What kW/rack now and later? Is peak summer the design point?
(4)	Grid/interconnect constrained buildout	MW availability/timeline, permitting	Demand reduction, phased deployment, storage/DR, procurement	Is interconnect delay the schedule driver? What MW cap is firm?
(5)	Carbon constrained operations	CO ₂ cap/accounting rules	Carbon-aware scheduling, flexible loads, procurement	Average vs marginal carbon accounting? Can batch loads shift in time?

While cooling and workload scheduling dominate the facility’s direct energy overhead, large AI training runs increasingly stress the intra-cluster fabric (all-reduce/all-to-all, checkpointing, data movement). As cluster size grows, performance can become fabric-bound, scenario (1), causing

idle GPUs and reduced scaling efficiency. That efficiency loss can dominate total GPU-hours even when the network is a smaller fraction of IT power. Photonics (e.g., more efficient optics, higher-radix fabrics, reconfigurable optical connectivity) is therefore modeled here as a “gain factor” on effective utilization rather than a simple watt-saver: improving the fabric can reduce required GPU-hours for a fixed training objective, which cascades into lower AI datacenter campus power demand and CO₂ emissions. The presented analysis provides a clean and transparent framework to quantify the impact of this lever.

Executive Summary: The result of this analysis under stated assumptions (see below) is shown in Fig. 1, signaling significant electricity cost savings for photonics interconnects/system behavior. Fig. 2 provides percent saved, MW saved, estimated equivalent GPU savings, and CO₂ emissions saved.



Baseline-Free Photonic Scaling Model

Disclaimer:

- This assessment provides a transparent, baseline-free sensitivity model that converts an assumed effective scaling gain into percent savings (MW, GPUs, CO₂).

This assessment does **not** provide a Jupiter-calibrated digital twin; in particular it does **not** claim interconnect is the dominant bottleneck for Jupiter.

Follow-on studies need to estimate effective scaling gain from workload traces, scheduler behavior, and fabric telemetry, here we assume for simplicity that it is a model input.

Core variable definitions:

Assumed inputs

- P_0 : baseline facility power per building [MW].
- PUE: power usage effectiveness [dimensionless].
- f_{GPU} : fraction of IT power attributable to GPUs [dimensionless, 0–1].

- W_{GPU} : Watts per GPU-equivalent [**W**].¹⁾
- $N_{\text{GPU},0}$: Baseline GPU equivalent.¹⁾
- $N_{\text{GPU},\text{after}}$: Capacity scaling.
- I : Carbon intensity [**lb CO₂/kWh**].
- C : C = electricity cost [**\$/kWh**].

Note:

¹⁾ GPU-equivalent = average board power under the assumed operating point (not nameplate TDP), used only for capacity accounting.

Photonic Lever for Scenario (1):

- g : effective scaling gain from improved interconnect/system behavior [**dimensionless**]. Interpretation: for the **same delivered workload**, the required capacity scales as $1/(1 + g)$. (see appendix for range selection)

If baseline effective utilization is u_0 and a systems intervention raises it to u_1 , then a capacity-equivalent gain is $g \approx u_1/u_0 - 1$ for workloads where lost utilization is predominantly due to coordination/communication limits.

Baseline-Free Savings

We model the “after” facility power as:

$$P_{\text{after}} = \frac{P_0}{1 + g} [\text{MW}]$$

Saved facility power:

$$P_{\text{saved}} = P_0 - P_{\text{after}} = P_0 \frac{g}{1 + g} [\text{MW}]$$

Saved fraction:

$$\text{Saved fraction} = \frac{g}{1 + g}$$

Mapping Facility MW → GPU-Equivalents

IT power:

$$P_{\text{IT}} = \frac{P_0}{\text{PUE}} [\text{MW}]$$

GPU power:

$$P_{\text{GPU}} = f_{\text{GPU}} P_{\text{IT}} [\text{MW}]$$

Baseline GPU-equivalents:

$$N_{\text{GPU},0} = \frac{P_{\text{GPU}} \cdot 10^6}{W_{\text{GPU}}} [\text{count}]$$

Capacity scaling:

$$N_{\text{GPU},\text{after}} = \frac{N_{\text{GPU},0}}{1 + g}; N_{\text{GPU},\text{saved}} = N_{\text{GPU},0} \frac{g}{1 + g}$$

Estimated Annual CO₂ Emission

Annual emissions (metric tonnes/year) for a constant carbon intensity I :

$$\text{CO2}_{\text{t/yr}} = \frac{P(\text{MW}) \cdot 1000 \cdot 8760 \cdot I}{2204.62}$$

Thus, CO₂ avoided under stated assumptions is:

$$\text{CO2}_{\text{saved}} = \text{CO2}(P_0) - \text{CO2}\left(\frac{P_0}{1+g}\right)$$

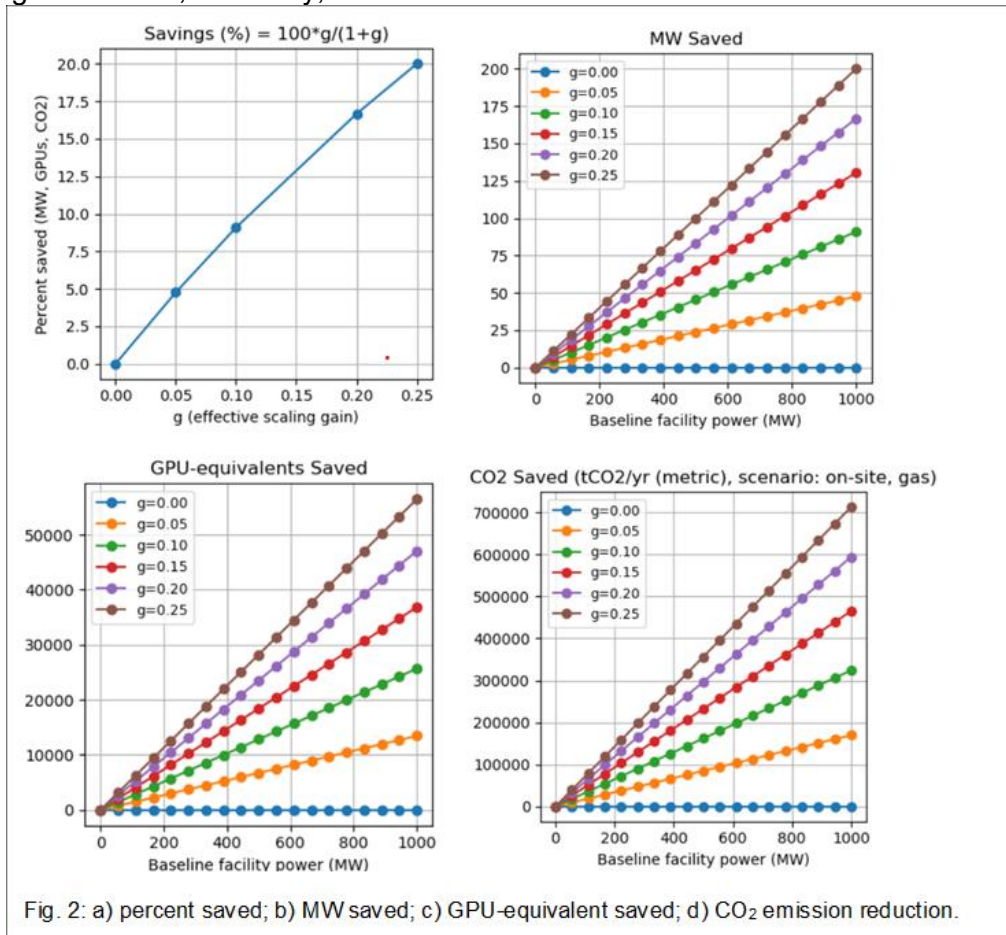
Estimated Annual Power Savings:

Annual cost/electricity savings,

$$\text{Annual Cost Savings} = P_{\text{saved}} \cdot 1000 \text{ kW/MW} \cdot 8760 \text{ hr/yr} \cdot C\$/\text{kWh}$$

Baseline assumption: $C = \$0.10/\text{kWh}$ (representative of industrial data center rates, which typically range \$0.05–0.15/kWh depending on region). Energy charges only.

Note: Total cost impact includes electricity savings shown here plus avoided capital expenditure (GPU hardware, networking equipment, racks) and reduced cooling infrastructure requirements. Fully-loaded cost per MW-year for hyperscale data centers typically ranges \$8M to 15M depending on location, efficiency, and contract structure.



This screening model is intentionally simple: it does *not* assert which bottleneck dominates Project Jupiter. It provides a transparent sensitivity baseline and a structured way to determine whether interconnect-driven scaling (g) is a material lever worth estimating via further analysis using workload traces, scheduler behavior, and fabric telemetry.

Appendix:

Literature based range selection for effective scaling gain (g):

$g = 0.00$ (**no gain**): Included as the null case because many deployments are compute-bound or already sufficiently provisioned on the fabric; in those regimes, interconnect changes may not materially alter effective throughput.

$g \approx 0.05$ (conservative, “small-but-real” gain): It is conservative relative to published results showing that mitigating communication contention can measurably increase GPU compute utilization in multi-tenant training environments. For example, **Crux** (SIGCOMM’24) reports 8.3%–14.8% GPU computation utilization improvement in a 96-GPU testbed and up to 23% improvement in trace-based simulation, by reducing contention among training jobs.

$g \approx 0.10$ (plausible mid-case if contention is nontrivial): 10% is within the scale of improvements reported by systems work that targets the communication/coordination inefficiencies of large training clusters (e.g., scheduling/communication-aware approaches like Crux). This value is used as a plausible sensitivity level when the workload exhibits meaningful collective/communication overhead and/or co-tenant interference.

$g = 0.20$: We include 0.20 primarily to show the stakes if the deployment is operating in a regime where data movement and contention are major bottlenecks and where mitigation recovers a significant fraction of lost throughput. This is consistent with the upper end of Crux’s reported utilization gains (up to ~23% in trace simulations). It is also well below extreme pathological cases documented in the literature—e.g., vClos reports that ECMP hash collisions in a 100 Gbps cluster can cause up to a 60% decrease in training throughput under certain collision scenarios.

We extended the range of effective scaling gain beyond $g = 0.20$ for visualization purposes.